

Order Management System

Week -4

1. Database Design

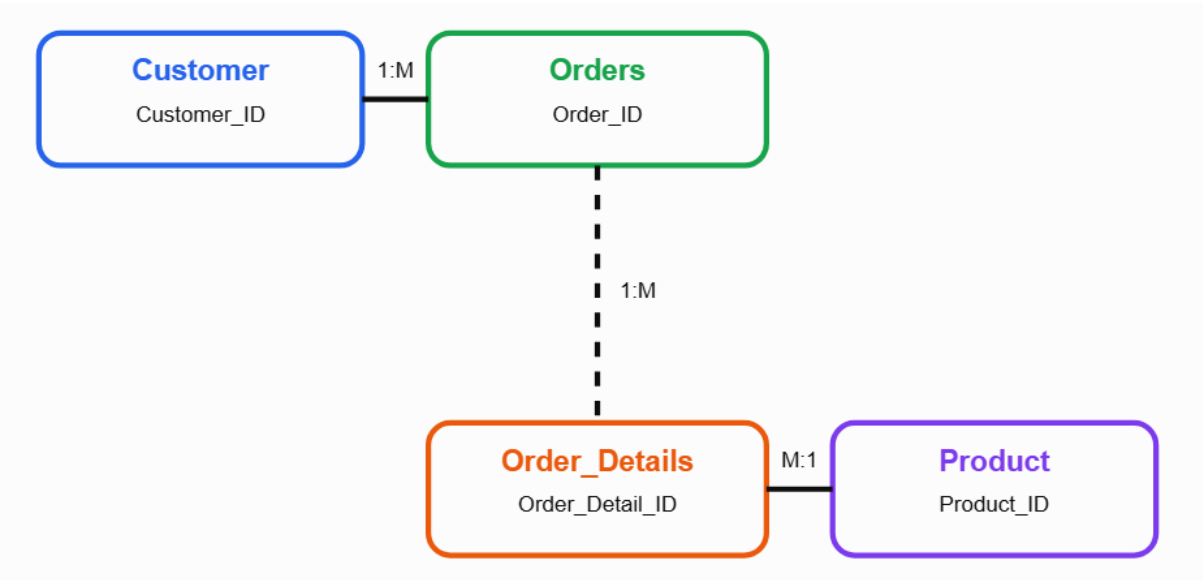
1.1 Orders Table

Attribute	Data Type	Constraint
Order_ID	INT	Primary Key
Customer_ID	INT	Foreign Key
Order_Date	DATE	NOT NULL
Total_Amount	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
Order_Detail_ID	INT	Primary Key
Order_ID	INT	Foreign Key
Product_ID	INT	Foreign Key
Quantity	INT	NOT NULL
Unit_Price	NUMBER(10,2)	NOT NULL
Subtotal	NUMBER(10,2)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Customer (  
2     Customer_ID INT PRIMARY KEY,  
3     Customer_Name VARCHAR2(100) NOT NULL,  
4     Email VARCHAR2(100) UNIQUE,  
5     Phone VARCHAR2(15),  
6     City VARCHAR2(50)  
7 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Orders (  
2     Order_ID INT PRIMARY KEY,  
3     Customer_ID INT,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2) NOT NULL,  
6     FOREIGN KEY (Customer_ID)  
7         REFERENCES Customer(Customer_ID)  
8 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO Customer VALUES
  2 (1, 'Ananya', 'ananya@gmail.com', '9876543201', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Customer VALUES
  2 (2, 'Rahul', 'rahul@gmail.com', '9876543202', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Customer VALUES
  2 (3, 'Priya', 'priya@gmail.com', '9876543203', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO Customer VALUES
  2 (4, 'Arjun', 'arjun@gmail.com', '9876543204', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO Customer VALUES
  2 (5, 'Meera', 'meera@gmail.com', '9876543205', 'Salem');

1 row created.
```

5. Order_Details Data Insertion

```

SQL> INSERT INTO Orders VALUES
  2 (1001,1,DATE '2026-09-20',2298);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1002,2,DATE '2026-09-21',999);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1003,3,DATE '2026-09-22',2499);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1004,4,DATE '2026-09-22',1599);

1 row created.

SQL>
SQL> INSERT INTO Orders VALUES
  2 (1005,5,DATE '2026-09-23',1798);

1 row created.

```

6. Order Modification Operations

6.1 Modify Order Amount

```

SQL> UPDATE Orders
  2 SET Total_Amount = 2398
  3 WHERE Order_ID = 1001;

1 row updated.

```

6.2 Modify Order Date

```

SQL> UPDATE Order_Details
  2 SET Quantity = 2,
  3     Subtotal = 1998
  4 WHERE Order_Detail_ID = 3;

1 row updated.

```

7. Order_Details Modification

```

SQL> UPDATE Orders
  2 SET Order_Date = DATE '2026-09-24'
  3 WHERE Order_ID = 1002;

1 row updated.

```

8. Customer Order History Report

```

SQL> SELECT
  2      c.Customer_Name,
  3      o.Order_ID,
  4      o.Order_Date,
  5      p.Product_Name,
  6      od.Quantity,
  7      od.Subtotal
  8 FROM Customer c
  9 JOIN Orders o
10 ON c.Customer_ID = o.Customer_ID
11 JOIN Order_Details od
12 ON o.Order_ID = od.Order_ID
13 JOIN Product p
14 ON od.Product_ID = p.Product_ID
15 ORDER BY o.Order_Date;

```

CUSTOMER_NAME

ORDER_ID ORDER_DAT

PRODUCT_NAME

QUANTITY SUBTOTAL

Ananya

1001 20-SEP-26

Women Floral Kurti

1 999

9. Customer-wise Order Summary

```
SQL> SELECT
  2      c.Customer_Name,
  3      COUNT(o.Order_ID) AS Total_Orders,
  4      SUM(o.Total_Amount) AS Total_Spent
  5  FROM Customer c
  6  JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_Name;
```

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_SPENT

Ananya	1	2398
Rahul	1	999
Priya	1	2499

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_SPENT

Arjun	1	1599
Meera	1	1798